

Real-Time Scheduling and Analysis of Fixed-Priority Tasks on a Basic Heterogeneous Architecture With Multiple CPUs and Many PEs

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Abstract—While accelerator-based heterogeneous architectures have gained traction in accelerating AI tasks, effectively managing them with stringent timing constraints remains a challenge. Although many scheduling and response time analysis approaches are proposed for multi-core or heterogeneous multi-core (i.e., big.LITTLE cores) processors, direct application of them to accelerator-based heterogeneous architectures with multiple CPUs and numerous processing units (PEs) often results in significant pessimism. This paper introduces real-time scheduling and comprehensive response time analysis from unit-level micro view to job-level macro view, for general accelerator-based heterogeneous architectures, greatly enhancing schedulability and utilization rates. We begin by establishing a general task execution pattern on heterogeneous architectures that integrates multiple CPU cores and various PEs. Subsequently, we present a real-time scheduling strategy and corresponding response time analysis based on this task execution pattern from micro to macro views. Through extensive experiments conducted on GEMM and AI workloads, our proposed scheduling and response time analysis significantly outperforms state-of-the-art scheduling algorithms, improving schedulability by 10.3% to 52.9%. Furthermore, experiments on NVIDIA GPU systems indicate a potential pessimism reduction of up to 30.7%. As we target general heterogeneous architectures, our approach can be readily applied to off-the-shelf accelerator-based heterogeneous computing systems, ensuring adherence to deadlines and enhancing schedulability.

Index Terms—Accelerator, CPU, GPU, heterogeneous computing, processing element, real-time scheduling, response time analysis.

I. INTRODUCTION

IN the realm of computing systems, spanning both embedded and cloud applications, there is a growing trend towards adopting an accelerator-based heterogeneous architecture. These

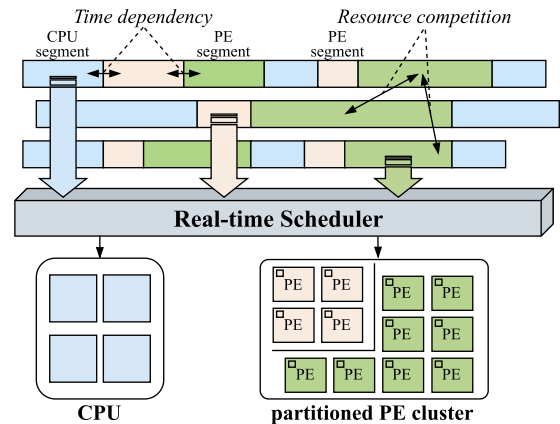


Fig. 1. Real-time scheduling of real-time tasks on the accelerator-based heterogeneous architecture.

platforms, exemplified by the likes of GPU servers [1], Xilinx UltraScale [2], and TI Keystone II [3], seamlessly integrate various types of processing units. These include CPU cores and parallel processing elements (PEs) such as GPU Streaming Multiprocessors or FPGA IP cores. This integrated approach leverages the unique strengths of each processor type, enabling enhanced performance and energy efficiency [4]. In numerous real-world applications, including fields such as autonomous driving [5] and robotics [6], the execution of AI tasks necessitates real-time responsiveness. This, in turn, places stringent demands on efficient task real-time scheduling and response time analysis to ensure compliance with tight timing constraints.

This work focuses on accelerator-based heterogeneous architectures, which integrate CPU processors with accelerators to handle diverse computational workloads in tasks. Tasks running on a basic accelerator-based heterogeneous computing platforms, which incorporate multiple CPUs and many PEs, typically have a segmented structure, as illustrated in Fig. 1. This heterogeneous architecture serves as the basis for System-on-Chip (SoC) designs and edge computing, while also acting as a foundational building block for distributed cluster systems. To optimize performance and energy efficiency, serial computation segments are usually allocated to CPU cores, while data-parallel segments are offloaded to PEs, which are known as GPU or FPGA segments. However, this interleaved execution pattern can cause dependencies and competition between parallel tasks, leading to complex scheduling challenges [7], [8]. The resulting

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